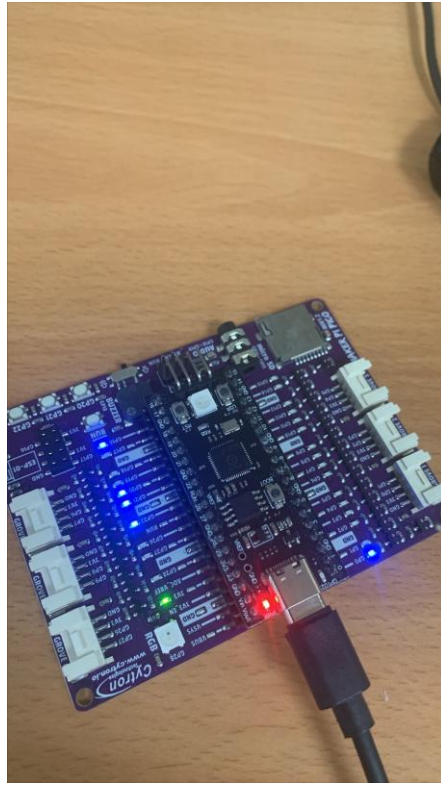
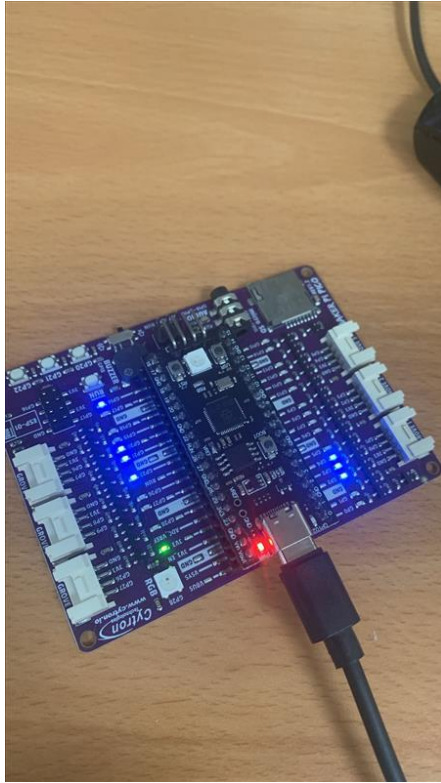


OS-6

U13157005 子二甲 吳翊弘



```
File Edit Selection View ... AssignmentQueue Update
EXPLORER
  ASSIGNMENTQUEUE
    .vscode
    build
    port
    src
      Agent.cpp
      Agent.h
      BlinkAgent.cpp
      BlinkAgent.h
      CMakeLists.txt
      CounterAgent.cpp
      CounterAgent.h
      main.cpp
      .gitignore
      CMakeLists.txt
      FreeRTOS_Kernel_import.cmake
      pico_sdk_import.cmake
  main.cpp 2
  Agent.cpp
  BlinkAgent.cpp
  CounterAgent.cpp
src > main.cpp > main(void)
1  /**
2   * Demo program to light 4 LEDs as binary random value
3   * Uses FreeRTOS Task
4   * Jon Durrant
5   * 15-Aug-2022
6   */
7
8
9  #include "pico/stdlib.h"
10
11  #include "FreeRTOS.h"
12  #include "task.h"
13  #include <stdio.h>
14  #include <stdlib.h>
15
16  #include "BlinkAgent.h"
17  #include "CounterAgent.h"
18
19  #define TASK_PRIORITY (tskIDLE_PRIORITY + 1UL)
20
21  // Blue LED (Pico onboard)
22  #define LED_PAD 25
23
24  // 8 LEDs
25  #define LED1_PAD 2
26  #define LED2_PAD 3
27  #define LED3_PAD 4
28  #define LED4_PAD 5
29  #define LED5_PAD 6
30  #define LED6_PAD 7
31  #define LED7_PAD 8
32  #define LED8_PAD 9
33
34  void mainTask(void *params){
35      BlinkAgent blink(LED_PAD);
36
37      CounterAgent counter(
38          LED1_PAD, LED2_PAD, LED3_PAD, LED4_PAD,
39          LED5_PAD, LED6_PAD, LED7_PAD, LED8_PAD
```

README

這個程式在 Raspberry Pi Pico 上執行 FreeRTOS 任務，藍色 LED 會持續以固定間隔閃爍，8 顆 LED 會接收主程式傳送的隨機數值並以二進位方式顯示，同時每隔一段時間更新並產生閃爍效果，整體透過多任務與 Queue 機制實現同步控制與持續運作。